

The Ephemeris

The Official Publication of the San Jose Astronomical Association

June 2025
Volume 36 Issue 2

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Sh2-223 and Sh2-224 by **Alex Ranous**. AP StarFire GTX 130mm on an AP 1100GTO mount. This is a two-panel mosaic through narrowband filters, totaling about 35 hours of exposures. <https://app.astrobin.com/i/mmz2wa>



The Whale Galaxy by **Francesco Meschia**. Over 24 hours of exposures through LRGBHa filters. Taken with an AT130EDT, AP Mach-1 GTO, and ZWO ASI294MM Pro camera. <https://app.astrobin.com/i/nnb5pr?r=M>



M88 by **Carl Svensson**. Part of a larger mosaic of the Markarian Chain. Taken with an SV80ST, EQ6-R Pro and LRGB filters through a ZWO ASI1600MM Pro camera. <https://app.astrobin.com/i/1h18lo>

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Calendar

BOARD MEETINGS + SPEAKER Saturday 6/7, 7/12, 8/9, 9/6 Houge Park, 6:00 - 7:30pm <i>Guest speaker follows meetings</i>	STARRY NIGHTS STAR PARTY Saturday 6/21, 7/19, 8/16, 9/13 RCDO, time varies
SOLAR SUNDAY & FIX-IT Sunday 7/6, 8/3, 9/7 Houge Park, 2:00 - 4:00pm	IMAGING SIG Tuesday 6/17, 7/15, 8/19, 9/16 Online, 7:30pm
IN-TOWN STAR PARTY Friday 6/20, 7/18, 8/19, 9/19 Houge Park, time varies	BINOCULAR STARGAZING Saturday 6/14, 7/19, 8/16 Coyote Valley, 9:30pm

Please refer to the SJ Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events) where you can find specific event times, locations, maps and possible cancellation due to weather.

SJAA Contacts

President	Larry Zurbrick
Vice President	Carl Svensson
Treasurer	Rob Jaworski
Secretary	Swami Nigam
Director 1	Larry Zurbrick
Director 2	Radhika Gupta
Director 3	Ken Miura
Director 4	Wolf Witt
Director 5	Swami Nigam
Director 6	Rob Jaworski
Director 7	Peter Melhus
Director 8	Glenn Newell
Director 9	Marianne Damon
Ephemeris Editor	Carl Svensson
Binocular Stargazing	Ed Wong
Fix-it Program	Larry Zurbrick
Hands on Imaging	–
Imaging SIG	Hy Murveit
Astronomy 101	Carl Svensson
In-Town Star Party	Muditha Kanchana
Library	David Kershaw
Loaner Program	Muditha Kanchana
Memberships	Kashaul Gangakhedkar
Publicity	Beth Johnson
Quick Start	Jonathon Lawton
Donations	Rich Gregor
Solar	Wolf Witt
Starry Nights	Joe Fragola
School Events	Jon Anderson
Refreshments	Marianne Damon
Social Manager	Beth Johnson
Speakers	Chris Gottbrath
Webmaster	Satish Vellanki
Volunteer Coordinator	Chris Gottbrath



SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly. Articles and Observation Reports for publication should be submitted by e-mail to ephemeris@sjaa.net. Please only submit original content (articles or photos) of your own authorship. Copyrighted content from other authors, including images, will be rejected.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

Member Programs

SJAA LIBRARY

SJAA offers another wonderful resource: a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may send all your questions/comments to librarian@sjaa.net.

TELESCOPE FIX-IT SESSION

Fix-It Day, sometimes called the Telescope Tune Up or the Telescope Fix-It program, is a really simple service the SJAA offers to members of the SJAA community for free, though it's priceless. Headed up by Larry Zurbrick, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

QUICK START PROGRAM

The Quick START Program, headed up by Jonathan Lawton, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick START sessions are generally held quarterly. The Quick START program has resumed under its new lead, Jonathon Lawton.

<http://www.sjaa.net/programs/quick-start/>

LOANER PROGRAM

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

ASTRO IMAGING SIG

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Hy Murveit, the Imaging SIG meets roughly every month over Zoom, often on the 3rd Tuesday evening of the month. We usually have a guest speaker presenting on some topic related to astro-imaging. Talks begin at 7:30pm. There is an opportunity to socialize and discuss imaging starting at 7pm when we start the Zoom call. Meeting topics, dates and times are posted on the SJAA Astro Imaging mail list in Google Groups. Past meetings are available on [YouTube](#).

<http://www.sjaa.net/programs/imaging-sig/>

ASTRO IMAGING WORKSHOPS AND FIELD CLINICS

Not to be confused with the Imaging SIG group, this newly organized program is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Keep an eye on the Google Groups and website for information about the next session.

Public Programs

ASTRONOMY 101

Astronomy 101 covers various topics related to beginner astronomy and star gazing. The popular Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by Carl Svensson. Content varies from "what's up in the night sky" to basics of star-gazing and telescopes to specific topics of interest. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>



SCHOOL STAR PARTY

The San Jose Astronomical Association conducts evening observing sessions (commonly called “star parties”) for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill. Contact ssp@sjaa.net for additional information.

<http://www.sjaa.net/programs/school-star-party/>

SOLAR OBSERVING

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time and location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

IN-TOWN STAR PARTY (ITSP)

This popular public star party is held every month, at our headquarters in Houge Park in San Jose. Enjoy celestial wonders from our very own backyard. Our volunteers are more than happy to show you the grate views through their telescopes. Check our Meetup page for specific dates and times. All are welcome!

BEGINNER ASTRONOMY FORUM

This exclusively online event is an opportunity for anyone to come and ask questions about the early stages of astronomy and star-gazing as a hobby. Unlike our other Astronomy 101 offers, this session is entirely guided by your questions, so come prepared! We will have two or three SJAA volunteers on-hand to share their knowledge.

STARRY NIGHTS

SJAA partners with the Santa Clara County Open Space Authority (OSA) to co-host monthly public star parties, called Starry Nights. These events are held at Rancho Canada del Oro (RCDO) Open Space Preserve just 30 minutes south of downtown San Jose. Come enjoy a night sky that is sheltered from city lights. Space is limited, so reserve your spot at:

<http://www.meetup.com/SJ-Astronomy/>

BINOCULAR STARGAZING

Want to learn the night sky? Did you know that you don't need to spend a ton of money on a telescope to do so? All you need is a decent pair of binoculars. Bring your 10x50mm binoculars, some comfy clothes, a chair, and join us on a tour of the night sky!

<http://www.sjaa.net/events/binocular-stargazing/>

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243
(Must be 18 years or older to apply)

New Membership or Renewal?

(check one)

☐

NEW

☐

RENEWAL

Membership Type

(check one)

☐

\$20 Membership with **online** Ephemeris

☐

\$30 Membership with **printed** Ephemeris

(mailed to address below)

NAME

ADDRESS

CITY/STATE/ZIP

PHONE

E-MAIL

Bring this form to any SJAA meeting or mail it to: SJAA, P.O. Box 28243 San Jose, CA 95159-8243

This newsletter is available online at <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net. Join or renew online at <http://www.sjaa.net/join-the-sjaa/>

Seasons of the Solar System

Kat Troche of the Astronomical Society of the Pacific via NASA's Night Sky Network

Here on Earth, we undergo a changing of seasons every three months. But what about the rest of the Solar System? What does a sunny day on Mars look like? How long would a winter on Neptune be? Let's take a tour of some other planets and ask ourselves what seasons might look like there.

MARTIAN AUTUMN

Although Mars and Earth have nearly identical axial tilts, a year on Mars lasts 687 Earth days (nearly 2 Earth years) due to its average distance of 142 million miles from the Sun, making it late autumn on the red planet. This distance and a thin atmosphere make it less than perfect sweater weather. A recent weather report from Gale Crater boasted a high of -18 degrees Fahrenheit for the week of May 20, 2025.

A diagram of Mars' orbit around the Sun, labeled with seasons for both hemispheres (Northern Spring/Southern Autumn, Northern Winter/Southern Summer, etc.). Key points like perihelion (closest to Sun) and aphelion (farthest from Sun) are marked, illustrating how seasons align with orbital position.

An artist's rendition of Mars' orbit around the Sun, and its seasons. Credit: NASA/JPL-Caltech

SEVEN YEARS OF SUMMER

Saturn has a 27-degree tilt, very similar to the 25-degree tilt of Mars and the 23-degree tilt of Earth. But that is

where the similarities end. With a 29-year orbit, a single season on the ringed planet lasts seven years. While we can't experience a Saturnian season, we can observe a ring plane crossing here on Earth instead. The most recent plane crossing took place in March 2025, allowing us to see Saturn's rings 'disappear' from view.

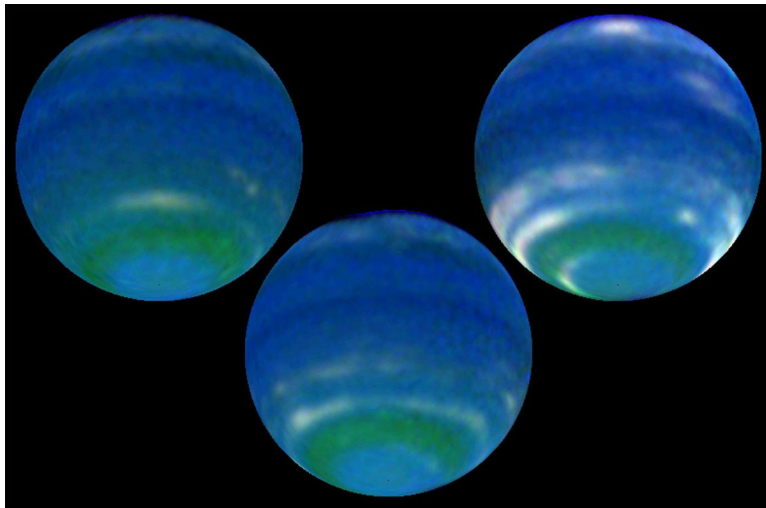
A LIFETIME OF SPRING

An image showing seasonal changes in Neptune between 1996 and 2002. The planet appears as a blue sphere with faint green and white atmospheric bands.

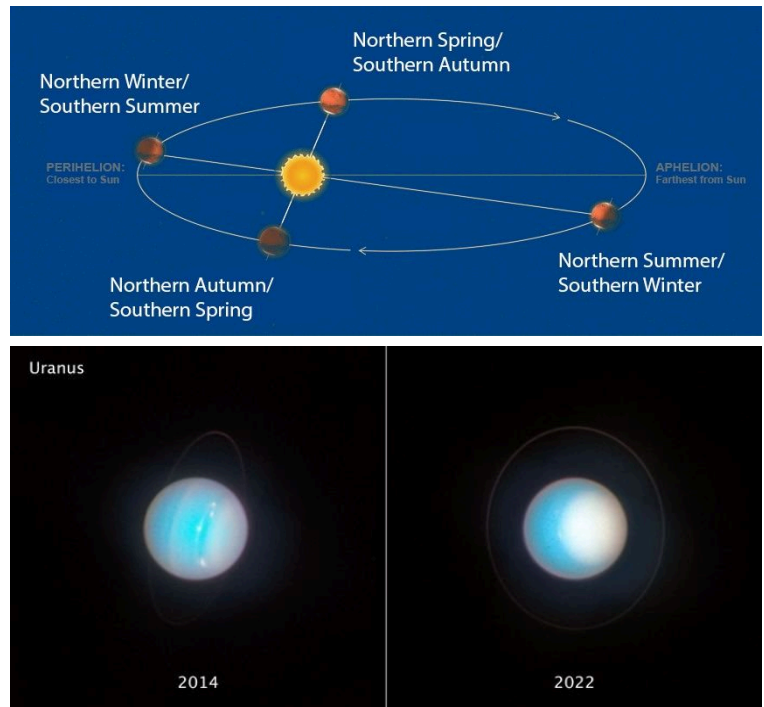
NASA Hubble Space Telescope observations in August 2002 show that Neptune's brightness has increased significantly since 1996. The rise is due to an increase in the amount of clouds observed in the planet's southern hemisphere. Credit: NASA, L. Sromovsky, and P. Fry (University of Wisconsin-Madison)

Even further away from the Sun, each season on Neptune lasts over 40 years. Although changes are slower and less dramatic than on Earth, scientists have observed seasonal activity in Neptune's atmosphere. These images were taken between 1996 and 2002 with the Hubble Space Telescope, with brightness in the southern hemisphere indicating seasonal change.

As we welcome summer here on Earth, you can build a Suntrack model (<https://solar-center.stanford.edu/AO/Sun-Track-Model.pdf>) that helps demonstrate the path the Sun takes through the sky during the seasons. You can find even more fun activities and resources like this model on NASA's Wavelength and Energy activity.



Top: Uranus rolls on its side with an 84-year orbit and a tilt just 8° off its orbital plane. Credit: NASA, ESA, STScI, Amy Simon (NASA-GSFC), Michael Wong (UC Berkeley); Image Processing: Joseph DePasquale (STScI). Right Top: An artist's rendition of Mars' orbit around the Sun, and its seasons. Credit: NASA/JPL-Caltech. Right Bottom: NASA Hubble Space Telescope observations in August 2002 show that Neptune's brightness has increased significantly since 1996. Credit: NASA, L. Sromovsky, and P. Fry (University of Wisconsin-Madison)



Chabot Astrophotography

SJAA members and astrophotographers Francesco Meschia and Imran Badr now have their astrophotography on display at the Chabot Space and Science Center as part of a new exhibit. It kicks off the grand re-opening of the observation deck at Chabot, and is truly a wonderful exhibit. It runs through the summer, so be sure to stop in and check it out! Francesco announced this honor to the observer's group earlier this year:

Imran and I are thrilled to share that we've been invited to exhibit our astrophotography in an upcoming display at the Chabot Space & Science Center in Oakland, alongside a stellar group of Bay Area astrophotographers, including Rogelio Bernal Andreo, Stuart Forman, Philippe Fossier, Brian Fulda, Gary Lopez, Bob Minor, Tara Mostofi, Judy Schmidt, and Jared Wilson.

We feel honored to be part of this exhibition. Developed by Chabot in partnership with the Space Telescope Science Institute, this display coincides with the Hubble Space Telescope's 35th anniversary, and it aims to provide visitors with an engaging entry point into astronomy, astrophotography, and space science.



Imran's rendition of the Andromeda Galaxy on display at Chabot.



Francesco's rendering of M33 on display at Chabot.

My Twinkle Twinkle

Jane Taylor of England wrote the poem The Star, which was published in 1806:

Twinkle, twinkle little star,
How I wonder what you are!
Up above the world so high,
Like a diamond in the sky.

When the blazing sun is gone,
When he nothing shines upon,
Then you show your little light,
Twinkle, twinkle through the night.

Then the traveller in the dark
Thanks you for your tiny spark;
He could not see where to go,
If you did not twinkle so.

In the dark blue sky you keep,
And often through my curtains peep,
For you never shut your eye
Till the sun is in the sky.

As your bright and tiny spark
Lights the traveller in the dark,
Though I know not what you are,
Twinkle, twinkle, little star.

When my grandson was born, I decided it was time to dispel the mystery:

The star answers:
I'm a ball of hydrogen,
Fusing into helium!
My light is steady as can be,
But air currents make the twinkle you see,
I am very big and bright,
But so far away I'm just seen at night.

And our Sun answers:
I'm a ball of hydrogen,
Fusing into helium.
Ninety-three million miles high,
Lighting up your daytime sky,
I send you rays of heat and light
And when Earth turns away, it's night.

Twinkle, twinkle little star, now I know just what you are!

*Marianne Damon,
as presented to SJAA, June 28, 2015*

Observing Highlights

06/21

June Solstice

The summer solstice will occur at 02:40 UTC when the north pole is tilted towards the sun.

07/04

Mercury at Greatest Elongation

The planet Mercury reaches greatest eastern elongation of 25.9 degrees from the Sun. This is the best time to view it.

07/28

Delta Aquarids Meteor Shower

An average shower that can produce up to 20 meteors per hour at its peak, produced by cometary debris.

08/12

Perseids Meteor Shower

One of the best meteor showers, with up to 60 meteors per hour at its peak produced by comet Swift-Tuttle.

08/19

Mercury at Greatest Elongation

The planet Mercury reaches greatest western elongation of 18.6 degrees from the Sun. This is the best time to view it.

09/07

Total Lunar Eclipse

A total lunar eclipse occurs when the Moon passes completely through the Earth's dark shadow, or umbra.

New Moon: 6/25, 7/24, 8/23, 9/21

Full Moon: 6/11, 7/10, 8/9, 9/7

Source: <http://www.seasky.org/astronomy/astronomy-calendar-2025.html>

KIDS CORNER

Word Search

- ▶ The Sun
- ▶ Photosphere
- ▶ Chromosphere
- ▶ Sun Spots
- ▶ Solar Flare
- ▶ Corona
- ▶ Eclipse
- ▶ Aurora
- ▶ Sunspots

V S G J S L G V T B E Q P B D
R U P B E B H B D H E I V V K
A N H A V X C J V X C G I D O
U S O H R I H R U T I Y L A A
R P T U T H R H S Z Q Y O S R
O O O M U S O L A R F L A R E
R T S S H E M Q T J Y S Q O C
A S P T U Z O S Y H S U J E O
R X H W D D S S M S E N M V R
U V E M O M P C J E W S W U O
S W R Q Z U H S H U V P U Y N
L X E C J J E E D Q U O V N A
Y P A A D V R D X V U T E Q P
G F L J X S E C N I B S U D T
D K V V O B A E C L I P S E E

Fun Facts

- ▶ The Sun's gravity holds the entire Solar System together!
- ▶ The sun is about 100 times wider than Earth.
- ▶ The Sun doesn't have a solid surface like Earth and the other rocky planets and moons.
- ▶ The part of the Sun commonly called its surface is the photosphere.
- ▶ The photosphere is about 10,000 degrees Fahrenheit.

Source: science.nasa.gov

Summer Space and Science

Member and contributor, L.G. Boyll, has a list of great science and space-themed summer adventures for you to check out.

CALIFORNIA ACADEMY OF SCIENCES - SAN FRANCISCO, CA

<https://www.calacademy.org/>

With a natural history museum, aquarium, multi-story rainforest, and planetarium, you won't run out of things to check out at the Cal Academy of Science. The Morrison Planetarium hosts a 75-foot dome that will immerse you inside cutting edge visualizations and hyper-realistic virtual environments. 'Incoming!' which is narrated by George Takei takes you up close with asteroids and comets while exploring the past, present and future of our Solar System. For adults, you can check out their Nightlife (advance tickets required) that happens on Thursdays and hosts a new theme with interactive exhibits, cocktails and music.

CHABOT SPACE & SCIENCE CENTER - OAKLAND, CA

<https://chabotspace.org/>

Located in the picturesque Redwood Regional Park of Oakland, the Chabot Space & Science Center is a non-profit science center that is home to the NASA Ames Visitor Center. It hosts an observatory, planetarium, exhibition galleries and more! Current exhibitions explore the mysteries of light and the Sun. You can pilot a rover mission control, see an authentic 7K-OK Russian Soyuz Descent Module, and even build your own test design to see how it stands up to other worlds. You can also register for summer events like family nature adventures or a celestial-themed paint and sip.

And if you can't make the trek to Oakland, you can still check out the Ames Gift Shop in Silicon Valley in Mountain View which has unique NASA gifts and official NASA gear.

CHILDREN'S DISCOVERY MUSEUM OF SAN JOSÉ - SAN JOSÉ, CA

<https://www.cdm.org>

Explore, test, tinker, and create with over 150 interactive exhibits and programs in the sciences, humanities, arts, nature, and health. This award-winning museum sparks learning through play and provides endless hours of family fun, whether you're creating a ginormous bubble, manipulating objects with air pressure, igniting scientific inquiry, or splashing together while sending a ball soaring atop a jet of water.

FREMONT PEAK OBSERVATORY ASSOCIATION - SAN JUAN BAUTISTA, CA

<https://fpoa.net/>

Located in Fremont Peak State Park near San Juan Bautista, Fremont Peak Observatory offers evening public programs that include an informative media presentation by a volunteer interpreter in the observatory classroom. When dark enough and weather permitting, the public is invited to observe with the 30" telescope and other telescopes until at least midnight. Daytime solar programs run from 2 to 5 PM and use special filters that allow you to safely view details on the sun's surface.

LICK OBSERVATORY - MT. HAMILTON, CA

<https://www.lickobservatory.org/>

If you haven't yet had the pleasure of visiting, you're probably still familiar with seeing the prominent white buildings that sit atop Mt. Hamilton. At 4,200 feet and only 20 miles from San José, the historic Lick Observatory is an iconic part of our skyline. A photogenic spot - it's a great destination for a day or night time visit to learn more about Lick Observatory's research and instruments. No reservations are needed for the free weekend talks inside the dome of the historic 36-inch Great Refractor! You can also check out their visitor center and gift shop to learn more about their history and astronomy-themed merchandise. Or register ahead for one of their many other events that include public evening tours, concerts with astronomy lectures, and photography nights! Make sure you allow at least one hour from San José. The road up to the Observatory is scenic but winding!

THE TECH INTERACTIVE - SAN JOSÉ, CA

<https://www.thetech.org/>

The Tech Interactive's galleries, which offer hands-on opportunities to explore how technological innovation is changing our lives and helping us tackle big challenges. Whether you have an hour or a full day, our exhibits will inspire, educate, and entertain. Make sure to check out the Space Exploration exhibit where you can maneuver a jet pack chair to fix a satellite, explore a futuristic Martian habitat, control a rover, conduct a scientific mission and even touch a piece of Mars!

SJAA By The Numbers

March 2025 - May 2025

Membership	456	(+15)
X Followers	908	(+7)
Facebook Followers	838	(+2)
Blue Sky Followers	552	(+108)
Instagram Followers	185	(+16)

The View from Below the Skies



STAR PARTIES HAVE EVOLVED

since I began attending (and volunteering at) so many decades ago. Amateur astronomers that shared their telescopes with the public in the distant past pointed their instruments mostly by manual means. Star hopping and star charts were the “de rigueur” of the day. Some tele-

scope operators had encoders and a connected microcontroller database setup that allowed them to “push to” a given celestial object. I was always impressed by my fellow amateur astronomers that could star hop to any Messier or NGC object in much less than a minute and just explain that it was no big deal. They would tell me you just have to look for this so and so asterism and move two and a half fields of view in such and such direction! It always worked in relatively dark sky locations. Being able to look through the eyepiece, seeing a dim object with averted vision (averted imagination?!) and realizing what one was observing was enough for most people. I only heard, “Oh, another dim fuzzy” occasionally!

But times have changed. Most telescopes that I see at star parties are automated in that they are attached to go to mounts. Just enter the object you want to observe and wait

a few seconds. Not necessarily a bad thing since the public can see more objects and spend less time waiting for the telescope to be pointed. More time is spent explaining to the public what they are observing. I still hear the “Oh wow” from folks visually observing planets like Jupiter and Saturn and the brighter Messier objects. Interacting with the public is why many of us enjoy volunteering at star parties.

Visual astronomy now appears to be going by the wayside at star parties with the advent of Electronically Assisted Astronomy (EAA) telescopes. No longer do we need to explain why the view through the eyepiece does not appear like a Hubble telescope image. It is not because there is no eyepiece in the case of EAA telescopes. We can show dimmer objects on a tablet thanks to automatic image stacking. The public can see color and much more detail in the images and Bortle 8 light polluted skies are less of an issue. Thankfully our hobby has evolved and we can share more interesting celestial wonders with the public! No more “Oh, another dim fuzzy” hopefully will be uttered. Ad astra.

Larry Zurbrik
President, SJAA

Board Minutes

MARCH, 2025

REIMBURSEMENT AND BUDGET The following expenditures were approved by the board: \$99 for Election Buddy (Wolf), \$116.99 for mount repair (Wolf), \$82 for food for volunteers at the Santa Cruz STEAM Expo (Rob), \$275 for Game Wheel prizes and first aid kit (Wolf), \$100 for tabletop signs for QR code-based sign-in.

STARS OVER YOSEMITE SJAA has the opportunity to volunteer at a summer star party in Yosemite in July. A number of volunteers have been identified, but logistics about parking, transportation, and camping are still being worked out.

PARTNERING WITH OTHER ASTRO CLUBS Wolf has been in contact with Bill O’Neil of the Seattle Astronomical Society as well as the Santa Cruz Astronomy Club with the goal of cross-promoting our events through these other clubs, and vice-versa. Wolf will follow up with these organizations, and Rob will follow up with MIRA.

APRIL, 2025

VOLUNTEER MEET-AND-GREET Wolf proposed having a team of leaders available to meet with prospective volunteers on short notice, in order to keep them engaged and make the best use of their energy. This meet-and-greet would be done for every volunteer, unless they

express interest in specific programs. Wolf, Swami, Larry, and Marianne offered to staff this effort.

DISCORD FOR SJAA Carl came ready with a rollout plan, but encountered some resistance. Swami suggested trialing Discord with program leads. Carl once again provided the server information.

MAY, 2025

REIMBURSEMENT AND BUDGET The following expenditures were approved by the board: \$260.88 for parts for astro image sharing stations and a step ladder, \$530 conditionally approved for a video camera for speaker and astro 101 presentations.

MEMBER SURVEY Results from a proposed membership survey are rolling in. Initial responses from over 50 members are giving us insight into the awareness, utilization, and satisfaction with various offerings from SJAA.

MEMBERSHIP DATABASE Carl has been proposing a web application to replace our current spreadsheet-based tracking of memberships. A demo was given in April, and Rob provided some feedback and questions that were discussed over Discord. Carl asked for a vote, but there were more questions raised regarding the choice of technology. More discussion will be held offline.

Excerpts from this quarter’s board meetings are above. Full minutes are available online at <https://www.sjaa.net/about/board-meeting-minutes/>

How To Get Involved and Stay in Touch

Did you know that SJAA is on Social Media!? Of course we are. We don't live under a rock, afterall. If you just can't get enough astronomy, consider following us on any one of our social media platforms:

- X (Twitter) [@sj_astronomy](#)
- Facebook [@sjastronomy](#)
- Instagram [@sjastronomy](#)
- Blue Sky [@sjastronomy](#)
- Astrodon [@sjastronomy](#)

Not a fan of social media? No problem. Your SJAA membership gives you access to several important Google Groups. These groups serve both as forums and email lists, and are vital for club communications. Every member is added to the [members@sjaa.net](#) group, where SJAA leadership will make announcements about timely or important events. This group is read-only for most members. If you are a member and aren't getting messages, reach out to us for help at [AskSJAA@sjaa.net](#).

Two other Google Groups are vital to round out your membership experience: [sjaa-observers@googlegroups.com](#) and [sjaa-astroimaging@googlegroups.com](#). The observers group is great for connecting with other members about current astro events and member-only observing opportunities. The imaging group is for all our members with special interest in and astroimaging. Share

your photos and get help from expert astrophotographers amongst our ranks. Both of these forums are accessible from [groups.google.com](#) as well as via e-mail. We encourage all members to read and post to these groups!

And then there's Meetup. Love it or hate it, SJAA has a robust presence on Meetup. This is where we post all of our public events. Many of our events are open to the public, and that includes SJAA members! Be sure to check our meetup page for notifications and up-to-date information about public events. Since many of our events are dependent on weather, it is important to check for updates close to the event to make sure it is not cancelled. For public events, this happens on Meetup. Find our page at: <https://www.meetup.com/sj-astronomy/>.

Finally, our website is a great resource as well. Our full calendar of events is featured on the front page, and additional celestial calendars are also available. Our website has details about all of our member programs, including information about how to take advantage of all the club has to offer, including the QuickSTART and Loaner programs.

We have tons of resources for our members and the public alike. If you have any trouble finding them, just reach out to us at [AskSJAA@sjaa.net](#). We will be happy to point you in the right direction. Until then, clear skies!

Brief Updates

OFFICER ELECTIONS

Officer elections (using ElectionBuddy electronic voting) kicked off on Sunday 3/9/2025 and concluded on Saturday 3/15/2025. Wolf (director4) conducted the election. There was exactly one candidate for each position - the current incumbent. Only the directors were eligible to vote in this election. Voter turnout was 78% - 7 out of 9 directors voted. All the 4 incumbents were reelected to their positions for a 1 year term ending March 2026. Congratulations to the 2025 officer team:

- **President** Larry Zurbrick
- **Vice President** Carl Svensson
- **Secretary** Swami Nigam
- **Treasurer** Rob Jaworski

OUR HISTORY

Thanks to Robert Garfinkle, we are able to restore some of the gaps in SJAA's history! On our website we have copies of the Ephemeris from 1999 to present. Bob heard our call for any earlier editions and came through with 1991 on. Over the next few months we will scan and post them on our website. Thank you, Bob!

COVER PHOTOS

The photo gracing our front cover this month should be a target that is familiar to any Hubble Space Telescope fans. It's the Eagle Nebula, or M16. Captured in stunning details and vibrant, narrowband colors, photographer Imran Badr composed this image out of nearly thirty hours of exposures with his Celestron EdgeHD 9.25" telescope and ZWO ASI2600MM Pro camera. More info on Astrobin: <https://app.astrobin.com/i/tkz695>

This month's back cover photo comes to us from Alex Ranous, and features an oft-overlooked object in Orion: SH2-263, or the "Raspberry Nebula." Taken with his Astro-Physics StarFire GTX 130mm telescope on an 100GTO-AE from the same brand, this combination reflection and emission nebula shines just "off Orion's shoulder," as Alex puts it. More info on Astrobin: <https://app.astrobin.com/i/s4021j>

If you would like to submit your images, articles, jokes, poems, observation reports, or eye-witness accounts to The Ephemeris, please check out our submission guidelines on our website: <https://www.sjaa.net/about/sjaa-newsletter-ephemeris/>

